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Graduation Project 1 Report

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الخلاصة

أدى النمو السريع لأجهزة إنترنت الأشياء (IoT) إلى زيادة مخاطر الهجمات الإلكترونية، وخاصة هجمات برامج الفدية. تعاني العديد من أجهزة إنترنت الأشياء من محدودية آليات الأمان والموارد الحاسوبية، مما يجعل اكتشاف برامج الفدية أكثر صعوبة مقارنةً بأنظمة الحاسوب التقليدية. غالبًا ما تعتمد أساليب الكشف الحالية إما على مراقبة الشبكة أو مراقبة الأجهزة المضيفة فقط، مما قد يقلل من قدرة الكشف الإجمالية.

يبحث هذا المشروع في اكتشاف برامج الفدية في بيئات إنترنت الأشياء باستخدام ثلاثة أساليب مختلفة للتعلم الآلي: الكشف القائم على الشبكة، والكشف القائم على الأجهزة المضيفة، والكشف الهجين. تستخدم الدراسة مجموعة بيانات ToN-IoT، التي تحتوي على بيانات حركة مرور الشبكة وبيانات سلوك النظام المستندة إلى القياس عن بُعد. تم تطبيق خمسة نماذج للتعلم الآلي وتقييمها، وهي: الغابة العشوائية (Random Forest)، و XGBoost، و CatBoost، والانحدار اللوجستي (Logistic Regression)، وخوارزمية أقرب الجيران (KNN).

أُجريت التجارب باستخدام نفس خطوات المعالجة المسبقة، ومعايير التقييم، واستراتيجية التحقق لضمان مقارنة عادلة بين جميع الأساليب. أظهرت النتائج أن نماذج التعلم الجماعي حققت أفضل أداء إجمالي في معظم التجارب. في النهج القائم على الشبكة، حققت نماذج Random Forest و XGBoost و CatBoost نتائج كشف عالية جدًا، حيث بلغت دقة الكشف 1.00 وقيم F1 قريبة من 0.99. أما في النهج الهجين القائم على مستوى الميزات، فقد حقق نموذجا XGBoost و CatBoost أعلى أداء إجمالي بين جميع النماذج التي تم تقييمها.

تُبين نتائج هذه الدراسة أن دمج المعلومات المستندة إلى المضيف والمعلومات المستندة إلى الشبكة يُحسن أداء كشف برامج الفدية في بيئات إنترنت الأشياء. كما تُوضح النتائج فعالية تقنيات التعلم الجماعي في كشف سلوك برامج الفدية من منظورات كشف مختلفة.

Abstract

The rapid growth of Internet of Things (IoT) devices has increased the risk of cyberattacks, especially ransomware attacks. Many IoT devices have limited security mechanisms and limited computational resources, which makes ransomware detection more difficult compared to traditional computer systems. Existing detection approaches often depend on either network-based monitoring or host-based monitoring only, which may reduce the overall detection capability.

This project investigates ransomware detection in IoT environments using three different Machine Learning (ML) approaches: network-based detection, host-based detection, and hybrid detection. The study uses the ToN-IoT dataset, which contains both network traffic data and telemetry-based system behavior data. Five ML models were implemented and evaluated, including Random Forest (RF), XGBoost, CatBoost, Logistic Regression (LR), and K-Nearest Neighbors (KNN).

The experiments were conducted using the same preprocessing steps, evaluation metrics, and validation strategy to ensure fair comparison between all approaches. The obtained results showed that ensemble learning models achieved the best overall performance in most experiments. In the network-based approach, RF, XGBoost, and CatBoost achieved very high detection results, with accuracy values reaching 1.00 and F1-scores close to 0.99 for

ransomware detection. In the hybrid feature-level approach, XGBoost and CatBoost achieved the highest overall performance among all evaluated models.

The results of this study show that combining host-based and network-based information can improve ransomware detection performance in IoT environments. The findings also demonstrate the effectiveness of ensemble learning techniques for detecting ransomware behavior using different detection perspectives.

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List of Abbreviations

IoT	Internet of Things
ML	Machine Learning
DL	Deep Learning
RF	Random Forest
LR	Logistic Regression
KNN	K-Nearest Neighbors

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Chapter 1 – Introduction

1.1 Introduction

The increasing demand for IoT in our daily lives has introduced several security challenges that require attention, especially as these devices are now widely used in healthcare, smart homes, manufacturing, and other critical sectors [1][2]. Many IoT systems still lack strong security controls, which makes them vulnerable to various cyberattacks.

Ransomware is formally defined by the National Institute of Standards and Technology (NIST) as “Ransomware is a type of malicious attack where attackers encrypt an organization’s data and demand payment to restore access.” [3]. In practice, ransomware often enters devices through phishing emails, unsafe applications, or compromised websites. Once activated, it spreads quietly and encrypts files, preventing users from accessing their data. Attackers may also threaten to leak sensitive information unless a cryptocurrency payment is made within a strict deadline [1][4].

Recent studies show that ransomware attacks are more widespread than official reports suggest. One study estimated more than 350 ransomware incidents targeting Dutch companies between 2019 and 2022, with nearly 59% of the cases going unreported [5]. Small companies were even more affected, with estimates of more than 2,700 attacks during the same period [5]. These findings highlight that ransomware is growing rapidly and often remains hidden, particularly in environments with weak security measures.

Traditional signature-based detection is becoming less effective as modern ransomware families increasingly use obfuscation and polymorphism to evade static signatures [6]. Because of these limitations, several studies have proposed behavioral and machine-learning-based detection techniques that analyze patterns such as file activity, resource usage, and system events. However, many of these models are still too heavy for resource-constrained IoT devices because they require high CPU and memory or rely mainly on network-level analysis rather than host-level behaviors [7][8].

These limitations and opportunities highlight the need for practical, low-overhead ransomware detection methods that can operate directly within IoT environments.

This project aims to investigate ransomware detection using network-based, host-based, and hybrid ML approaches in IoT environments.

Here, This project aims to investigate ransomware detection using network-based, host-based, and hybrid ML approaches in IoT environments.

Unlike previous studies that focused on a single detection perspective, this work evaluates network-based, host-based, and hybrid ransomware detection approaches within IoT environments.

The remainder of this chapter provides a problem definition, outlines the main aims and objectives of the project, presents the expected timeline, and concludes with a summary of the chapter.

1.2 Problem Definition

Recent security reports, including assessments from Kaspersky and Check Point, indicate a significant increase in ransomware activity targeting IoT ecosystems. This surge is primarily attributed to weak firmware security, outdated patch cycles, and constant network exposure [9][10]. This trend highlights an urgent need for detection mechanisms specifically designed for IoT hardware and network patterns.

The core problem addressed in this project is the absence of a ML model capable of identifying ransomware behavior using three different approaches: network-based, host-based, and hybrid. Existing signature-based tools rely on either network-level or host-level detection and often use heavy ML models that require computational and memory resources exceeding typical IoT device limits, resulting in delayed or reactive detection under real-world workloads [11].

Furthermore, several recent studies demonstrate that modern ransomware families can initiate rapid encryption activity, making it difficult for most existing host-based and network-based solutions to respond efficiently [26]. Combined with the limited CPU and RAM capacities of IoT nodes, this creates a critical gap where attacks may escalate before any defensive action can be taken.

1.2.1 Project Scope

This project focuses on ransomware detection in IoT environments using supervised ML techniques. The study is limited to ransomware-class attacks using selected subsets from the ToN-IoT datasets. Three detection approaches are evaluated, including network-based, host-

based, and hybrid detection. The experiments are conducted using offline evaluation and binary classification (normal vs ransomware).

The scope of this work does not include real-device deployment or containment mechanisms.

1.3 Aims, Objectives and Project Timeline

1.3.1 Aim

To design, implement, and evaluate a hybrid ML model for ransomware detection in IoT environments using host-based and network-based features.

1.3.2 Objectives

1. Explore ransomware detection in IoT environments using ML models.
2. Implement and evaluate three different detection approaches: network-based, host-based, and hybrid.
3. Compare the performance of multiple ML models, including RF, XGBoost, CatBoost, LR, and KNN.
4. Evaluate the proposed approaches using standard classification metrics such as accuracy, precision, recall, and F1-score.
5. Identify the strengths and limitations of each detection approach in terms of detection capability and suitability for IoT environments.
6. Examine whether combining host-based and network-based features through a hybrid approach can improve ransomware detection performance.

1.3.3 Project Timeline

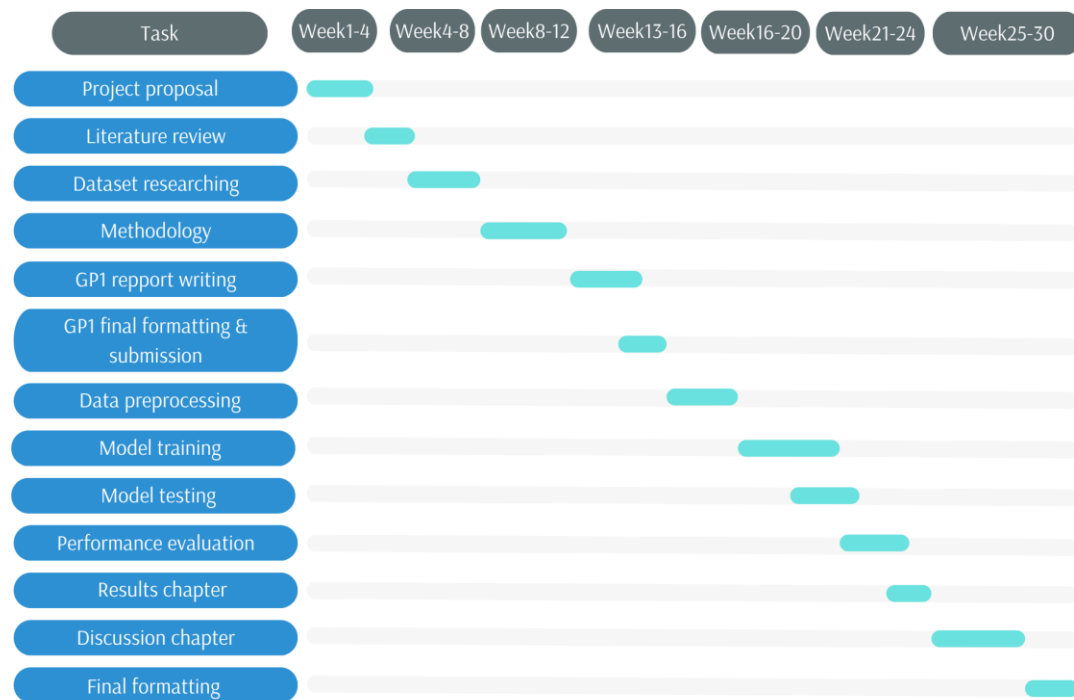


Figure 1: Project timeline showing the phases of the ransomware detection project across the academic year.

Chapter 2 - Literature Review Chapter

2.1 introduction

In the previous chapter, we discussed the problem definition, our aims and objectives for this project, and showcased our project timeline. In this chapter we will give a brief background about ransomware, present related works that are related to our project and explain how other researchers tried to detect ransomware in both computer and IoT environments. We summarize the main ideas, methods, and results used in these studies. After that, we compare their approaches so that we gain a better understanding of what has already been done and what still needs improvement. By doing this, we can gain a better vision of the gaps in previous studies.

2.2 Background

Ransomware is a type of malware that encrypts files or blocks access to a device until a payment is made. While ransomware attacks were originally focused on personal computers, they have recently expanded to IoT devices because these devices often have weak security, limited processing power, and they are always connected to the network, which makes them easier targets for attackers.

Their limited resources also mean that many traditional security techniques cannot be applied to them. For this reason, recent research has focused on finding new ways to detect ransomware in IoT environments, such as analyzing network traffic or using ML models. Thus, understanding how ransomware behaves in IoT environments is important for designing an effective detection model and for identifying which method would work the best.

This background illustrates that the problem lies not only in understanding ransomware, but also in the lack of a practical detection mechanism suitable for IoT environments. The limitations of IoT devices may reduce the practicality of computationally intensive ML and DL approaches, creating a significant challenge in ransomware detection for IoT systems..

2.3 Related Works

There are several recent research studies that focus on ransomware detection in both traditional computer systems and IoT environments. Each paper proposes a different method, model, or technique to improve detection accuracy and reduce false alarms. By analyzing these papers,

we aim to understand the strengths and limitations of existing approaches, identify the datasets and features they used, and highlight the challenges that they have not solved yet. A comparative summary of these studies is presented in *Table 1* to highlight the key differences and identify the specific gaps addressed by our proposed model. This review will help us determine the research gap and clarify why our proposed model is needed.

2.3.1 Summary of Reviewed Studies

Several previous studies have been conducted on ransomware and intrusion detection in IoT environments, employing a variety of methodologies. Sun et al. [12] proposed L-IDS, a multi-layer lightweight IDS which combines software checking with hardware assisted isolation using Trusted Execution Environments (TEE). In their application of their method with Naive Bayes they achieved an accuracy rate of about 86%, but the evaluation was limited to virtual simulation, i.e., pre-encryption behavior was not in the scope. Similarly, Shahid et al. [14] introduced CrossSentry, a cross-layer detection framework that enhances efficiency by analyzing logs from different IoT layers. Its evaluation was limited to a small number of devices and did not include analysis of polymorphic ransomware or large-scale deployments, but it showed high accuracy with no false positives.

For advanced modeling, Maniriho et al. [15] applied a Generative Adversarial Network (GAN) framework for polymorphic ransomware and achieved high accuracy (95-98%) with a DBN classifier. But the high computational overhead of this framework is not suitable for resource-constrained IoT devices. In contrast, Alrawi et al. [17] investigated monitoring at the hardware level using Hardware Performance Counters (HPCs) to ensure low overhead. The accuracy is mid 90s but it is not portable on various IoT platforms because of the hardware used.

Moreover, recent efforts have been directed at early detection and network level security. Hussain et al. [18] proposed an early-prediction model for healthcare IoT using GRU-TCN architectures to analyze API-call behavior before encryption. The model shows excellent prediction capabilities. But the applicability is limited to lightweight IoT logs as the model was trained on GPUs and used Windows API traces. In addition, Banset et al. [19] proposed an intrusion detection framework based on ML models trained on network traffic datasets like CICIDS2018. Despite the high accuracy of 98% for general cyber-attack prediction, the system lacks ransomware-specific analysis at the host level and requires cloud support, which limits its effectiveness for on-device detection in resource-constrained environments.

Table 1: Comparative Analysis of Existing Ransomware Detection Approaches

Study Name	L-IDS [12]	CrossSentry [14]	PRDL Polymorphic DL Framework [15]	RansomSentry [17]	GRU-TCN Early Prediction [18]	IoT-NIDS Prediction System [19]

Approach	Multi-layer IDS	Cross-layer	DL-based	HPC-based	API-based	Network-based
Dataset	Simulated IoT	IoT logs	Windows	IoT-like	Windows	CICIDS2018
Lightweight	Yes	Yes	No	Yes	No	No
Early Detection	No	Limited	No	No	Yes	No
Limitation	Tested on virtual devices only	Small-scale evaluation	Complex computation	Hardware-dependent	Not IoT-suitable	Not ransomware-specific
Reported Accuracy/ F1-score	87.1% / 86.1%	100%	0.962% / 98.6%	100%	96.98% / 97.02%	96.89% / 96.84%
Validating setting	Raspberry Pi simulation	Raspberry Pi device	Cuckoo Sandbox	Virtualized IoT (Raspberry Pi OS – Debian 12)	Windows API dataset	IoT-23 network flows

As shown in *Table 1*, existing ransomware detection studies still face several challenges related to IoT applicability and effective ransomware detection. Some approaches rely on complex deep-learning models or environment-specific features, while others are evaluated using simulated or Windows-based environments rather than real IoT scenarios.

In addition, several studies focus on general malware or network attacks without specifically targeting ransomware behavior in IoT systems. Other approaches depend on limited datasets or single-source behavioral information, which reduces their ability to capture ransomware activity from multiple perspectives.

These limitations highlight the need for an IoT-oriented ransomware detection approach capable of combining both network-level and host-level behavioral information to improve ransomware detection in resource-constrained IoT environments.

2.3.2 Dataset Overview

In previous studies on ransomware detection, several datasets have been used, including various types, such as system-based and network-based. The datasets differ in data type, feature representation, and suitability for IoT environments. Some datasets are focused on file level or PE header features; some datasets are focused on network traffic or behavioral telemetry. *Table 2* shows a comparison of most common ransomware datasets and their suitability for IoT devices.

Table 2: Comparative Overview of Ransomware Datasets Used in Previous Studies

Dataset Name (Year)	Data Type	Includes Ransomware?	Real IoT?	Suitable for IoT?
Ransomware detection dataset 2022 [20].	Host-Based	Yes – contains ransomware behavioral activity	No – collected from Windows-based systems	No – Windows-based behavioral dataset
RanSAP Dataset (2022) [25].	Host-Based	Yes – includes ransomware storage access patterns	No – collected from traditional computer systems	Partially – storage access behavior can support IoT-related analysis
Ransomware Dataset (2024) [27].	Host-Based	Yes – includes multiple ransomware families	No – based on Windows executable samples	No – focused on Windows ransomware samples
IoT-23: A labeled dataset with malicious and benign IoT network traffic. 2020 [31].	Network-Based	No – focuses on IoT malware and botnet traffic	Yes – collected from real IoT devices	Partially – real IoT traffic but does not include ransomware
ToN_IoT datasets. 2022 [37].	Hybrid (Network + Telemetry)	Yes – includes ransomware-related attacks	Yes – collected from real IoT devices	Partially – real IoT traffic but does not include ransomware

As shown in *Table 2*, many datasets used in previous ransomware detection studies are not fully suitable for IoT environments. Several datasets are designed for traditional Windows-

based systems rather than real IoT devices. In addition, some datasets focus on general malware or network attacks without including ransomware-specific behavior.

Furthermore, many existing datasets provide either host-based or network-based information only, which limits the ability to capture ransomware behavior from multiple perspectives. These limitations highlight the need for a hybrid IoT-oriented dataset and detection approach that combines both network and host-level behavioral information for more effective ransomware detection.

2.3.4 Research Gap

Based on the analysis of previous studies and datasets, several limitations can be identified in existing ransomware detection approaches. Many studies rely on a single source of evidence, such as network traffic or host-level behavior only, which limits the ability to capture complex ransomware activities [17][19]. In addition, several recent approaches utilize computationally intensive DL architectures that may not be suitable for resource-constrained IoT environments [15][18]. Furthermore, some studies were evaluated only in controlled or simulated environments with limited testing on real IoT devices [12][14][17].

Therefore, this project focuses on investigating network-based, host-based, and hybrid ML approaches within a unified evaluation framework to improve ransomware detection in IoT environments.

2.4 Conclusion

Many approaches rely on simulated environments or datasets that do not effectively represent real IoT behavior. In addition, several models are computationally intensive, which may reduce their suitability for resource-constrained IoT environments. Other works focus on a single detection source, such as network traffic or system behavior, which limits the overall detection capability.

Furthermore, the analysis of existing datasets shows that many are PC-based, lack IoT-specific characteristics, or do not combine multiple data sources. This reduces their effectiveness in capturing realistic ransomware behavior in IoT environments.

In conclusion, these limitations highlight the need for IoT-oriented ransomware detection approaches that integrate multiple sources of information to improve detection performance.

In the next chapter, the methodology used in this study will be presented, including the theoretical formalization and the experimental design.

Chapter 3: Methodology

3.1 Introduction

This chapter presents the methodology used in this study to detect ransomware attacks using ML techniques. After examining many literatures and based on the discussion in the previous chapter, three different detection approaches are considered: network-based, host-based, and hybrid.

Each approach provides a different perspective on ransomware detection, where network-based methods focus on traffic patterns, host-based methods analyze system behavior, and hybrid methods combine both to improve detection performance.

This chapter describes the theoretical framework, the ML models used, and the experimental design applied to evaluate these approaches. It also explains how the comparison between the approaches is conducted using consistent evaluation metrics.

3.2 Theoretical Formalization

3.2.1 Preliminaries

Ransomware detection in this study is considered a supervised classification problem. The objective is to classify each data sample as either normal behavior or a ransomware attack.

First, X is defined as the input feature space, where each sample contains a set of features describing system or network behavior. The corresponding label is represented as $y \in \{0,1\}$, where 0 represents normal activity and 1 represents ransomware.

A ML model is defined as a function $f(X)$ that maps input features to predicted labels. The model is trained using labeled data and then evaluated on unseen data to assess its ability to correctly classify ransomware attacks.

This formulation is consistently applied across all detection approaches considered in this study.

3.2.2 Formalizing Research Question

The main research question addressed in this study is:

How effectively can ransomware attacks be detected using Machine Learning models across different detection approaches (network-based, host-based, and hybrid)?

This question guides the design of the methodology and the evaluation process used to compare the different approaches.

3.2.3 Formalizing Machine Learning Framework

This study adopts a supervised ML framework for ransomware detection. In this setup, The implementation utilizes Python 3.13.9 as the primary programming language, leveraging the scikit-learn library for data preprocessing and model evaluation. Also, XGBoost and CatBoost libraries were used to implement advanced ensemble learning techniques.

Five different classifying models were selected and trained using labeled data consisting of input features and corresponding class labels:

- RF: Configured with `n_estimators=100` and a fixed `random_state=42`
- XGBoost: Configured with `n_estimators=100`, `max_depth=6`, and a `learning_rate=0.1`
- CatBoost: Implemented with `iterations=100`, `depth=6`, and a `learning_rate=0.1`
- LR: configured with `max_iter=1000` to ensure convergence
- KNN: Implemented with `n_neighbors=5`

All models were initialized with `random_state` of 42 to ensure reproducibility.

The workflow follows standard classification steps, including data preprocessing, feature selection, model training, and prediction. The same set of ML models is applied across all detection approaches to ensure consistency and fair comparison.

These selected models represent a balance between classification performance and model diversity. LR and KNN were included as baseline ML models, while ensemble methods such as RF, XGBoost, and CatBoost were selected for their ability to capture more complex behavioral patterns. All models were trained using the same training data and evaluated on unseen test data to measure their generalization performance. This setup supports a fair comparison aligned with the objective of evaluating ransomware detection approaches in IoT environments.

3.2.4 Formalizing Detection Approaches

This study investigates three main approaches for ransomware detection: network-based, host-based, and hybrid. Each approach utilizes the same ML framework but differs in the type of data and perspective used for detection.

The network-based approach focuses on analyzing network traffic data, while the host-based approach relies on system-level behavior. The hybrid approach combines both perspectives to enhance detection performance.

Each approach is implemented using the same classification models and evaluation metrics to ensure a fair comparison.

3.2.5 Analysis of Approaches

Each detection approach provides unique advantages and limitations. The network-based approach is efficient and suitable for real-time detection, but it has limited visibility into

internal system behavior. The host-based approach offers deeper insight into system activities, allowing for more detailed detection, but it may introduce higher computational overhead.

The hybrid approach combines both network and host perspectives, which has the potential to improve detection accuracy. However, this approach introduces additional complexity in data processing and model integration.

This analysis highlights the trade-offs between the approaches and provides a foundation for evaluating their performance in the experimental design.

3.2.6 Evaluation Metrics

The performance of the models is evaluated using standard classification metrics defined as follows:

Accuracy

To measure the overall correctness of the model, the accuracy metric is used, where a higher value indicates better detection performance.

In this context, TP is short for true positive, TN for true negative, FP stands for false positive, and FN for false negative.

$$Accuracy = \frac{(TP+TN)}{TP+TN+FP+FN} \quad (1)$$

Precision

This metric measures the proportion of correctly predicted malicious instances among all instances predicted as malicious.

$$precision = \frac{TP}{TP+FP} \quad (2)$$

Recall

Measures the ability of the model to correctly detect malicious instances.

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

F1-Score

The metric used to provide balance between precision and recall; this metric is useful when dealing with unbalanced datasets.

$$F1 = 2 \times \frac{precision \times recall}{precision + recall} \quad (4)$$

These metrics were selected to provide an overall evaluation of model performance. Accuracy reflects the total number of correctly classified instances among all predictions, while precision and recall are particularly important in ransomware detection, as correctly identifying malicious activity is critical. F1-score provides a balance measure between false positives and false negatives, making it suitable for evaluating models related to security applications.

3.3 Experimental Design

This section describes the experimental setup used to evaluate the performance of the proposed detection approaches.

The study utilizes the ToN-IoT dataset, which includes both network-based and host-based data. This dataset was selected because it provides a comprehensive representation of IoT environments and contains labeled records of normal and malicious activities, including ransomware-related behavior.

3.3.1 Data Splitting and Validation Strategy

Hold-out validation technique was applied to split the dataset into 80% for training and 20% for testing. Stratified sampling was used to preserve the original class proportions in both sets. We fixed the seed (`random_state=42`) in the split to ensure the reproducibility of the results.

3.3.2 Balance of Data and Pre-processing

To handle the inherent class imbalance of the ToN-IoT dataset, where the normal traffic is much larger than the ransomware samples, a random undersampling technique was applied to the training set. This assured that both classes were equally represented (1:1 ratio) during the model fitting phase.

The features were engineered by one-hot encoding the categorical variables and aligning the feature spaces of the training and testing sets to account for missing categories. Moreover, feature scaling was applied with `StandardScaler` to distance based and linear models (KNN and LR) and tree-based ensembles were trained on the unscaled feature set to preserve their natural decision boundaries.

The dataset enables the evaluation of different detection approaches from multiple perspectives, allowing a fair comparison between network-based, host-based, and hybrid models.

The independent variables in this study are the selected features extracted from the datasets, while the dependent variable is the classification label indicating normal or ransomware activity.

A set of ML models is used for evaluation, including RF, XGBoost, CatBoost, LR, KNN. The performance of these models is assessed using standard evaluation metrics, including accuracy, precision, recall, and F1-score.

To ensure a fair comparison, all approaches are evaluated using the same dataset, preprocessing steps, ML models, and evaluation metrics.

3.3.3 Network-Based Experimental Design

a. Preliminaries

Network-based ransomware detection depends on analyzing the traffic between network entities. Network traffic flow is defined as a sequence of packets that are transmitted between a source and a destination. These packets have attributes such as IP address, protocol type and ports.

The network dataset used here is a subset of the ToN-IoT dataset which is a connection-based, meaning that each record or row represents a network session that has multiple properties that allow the model to capture communication patterns at a higher level.

b. Features Definition

According to the ToN-IoT documentation [37], the following features were selected to represent network activity. These thirteen features were chosen from the initial set of over 40 attributes based on their low cardinality to prevent overfitting, their availability across all flow records, and their established relevance in prior ransomware detection studies [38].

Traffic Volume Features (Connection Activity):

src_bytes	Source bytes which originate from payload bytes of TCP sequence number.
dst_bytes	Destination bytes which are responded payload bytes from TCP sequence number.

duration	The time of the packet connections, which is estimated by subtracting ‘time of last packet seen’ and ‘time of first packet seen’.
src_port	Source ports which originate endpoint’s TCP/UDP ports.
dst_port	Destination ports which respond to endpoint’s TCP/UDP ports.
proto	Transport layer protocols of flow connections.
service	Dynamically detected protocols, such as DNS, HTTP and SSL.
conn_state	Various connection states, such as S0 (connection without reply), S1 (connection established), and REJ (connection attempt rejected)

Packet-Level Features (Statistical Activity)

src_pkts	Number of original packets which is estimated from source systems.
dst_pkts:	Number of destination packets which is estimated from destination systems.

DNS Features (DNS Activity)

dns_qtype	a numeric value which specifies the DNS query type.
dns_rcode	The response code values returned by the DNS server indicating the status of the query.
dns_rejected	DNS rejection, where the DNS queries are rejected by the server.

c. Feature Selection Justification

These features mainly rely on metadata extracted from network flows rather than deep packet inspection or payload-level analysis. In addition, the approach does not require payload analysis, HTTP content parsing, or SSL decryption, which simplifies the feature extraction process and supports faster traffic analysis.

This feature set contributes to faster processing time, and it is suitable for real-time detection in network monitored systems such as IDS and firewalls.

d. Formalizing the Network-Based Solution

The proposed approach follows a structured ML pipeline:

- 1- Data Merging: where multiple CSV network traffic files were combined in a single dataset.
- 2- Filtering: excluding any attack types that are not ransomware, after this step the only records left were either normal or ransomware ensuring a focused binary classification and reducing the dataset size to 854,944 records.
- 3- Feature Selection: 13 relevant features were selected to improve efficiency and reduce dimensionality.
- 4- Preprocessing: handling missing values, conversion of numerical features, and mapping categorical values.
- 5- Encoding: transferring the categorical variables using one-hot encoding.
- 6- Data Splitting: dividing the dataset into 80% training and 20% testing using stratified sampling.
- 7- Data Balancing: balancing the training set data using undersampling to ensure equal representation of both classes.
- 8- Scaling: scaling the features using standardization for models that require normalized input (ensemble models).
- 9- Model Training: Five ML models were trained: RF, XGBoost, CatBoost, LR, and KNN.
- 10- Prediction and Evaluation: Classifying the test data into normal or ransomware using the trained models.

e. Analysis of the Network-Based Approach

This approach is scalable for large network environments because it relies on network-flow features rather than deep packet inspection. In addition, the selected features support faster

traffic analysis and efficient processing of network activity. However, this approach has limited visibility into host-level behavior and may not effectively detect ransomware operating internally without significant network communication.

3.3.4 Host-Based Experimental Design

a. Preliminaries

The host-based approach focuses on analyzing system-level behavior within IoT devices to detect ransomware activity. Unlike network-based detection, which monitors external communication, the host-based approach examines internal system operations such as running processes, file system interactions, and memory usage.

Each data sample represents the behavior of the system over a specific time interval, capturing how applications interact with system resources. This includes monitoring process execution patterns, file modification activities, and system-level events. This approach provides deeper visibility into ransomware behavior, especially during early stages such as file encryption attempts.

b. Feature Concept

The host-based approach relies on extracting features that reflect system-level behavior. These features capture different aspects of system activity, including:

Process behavior: such as process execution patterns and resource consumption.

File activity: including file creation, deletion, and modification patterns.

Memory and system activity: representing system resource usage and abnormal behavior during execution.

These features are selected to represent behavioral patterns associated with ransomware, particularly those related to rapid file encryption and abnormal system activity.

c. Feature Justification

The selected host-based features were chosen due to their ability to capture early behavioral indicators of ransomware activity. Unlike signature-based detection, which relies on known patterns, these features reflect real-time system behavior, making them more effective in detecting unknown or evolving ransomware variants.

Additionally, host-based features provide detailed insight into internal system operations, allowing the detection model to identify malicious activities that may not be visible at the network level.

d. Formalizing the Host Solution

The host-based detection model follows a structured ML pipeline consisting of the following stages:

Data preprocessing: cleaning and preparing the dataset for analysis.

Feature extraction: selecting relevant host-based features representing system behavior.

Data splitting: dividing the dataset into training and testing sets.

Balancing and scaling: ensuring class balance and normalizing feature values.

Model training: applying ML algorithms to learn behavioral patterns.

Prediction: classifying instances as normal or malicious.

Evaluation: assessing model performance using standard classification metrics.

e. Analysis of the Host-Based Approach

The host-based approach provides detailed visibility into system-level behavior, allowing the detection model to identify ransomware activities such as abnormal file modifications and process execution patterns. This makes it highly effective in detecting ransomware during early stages of the attack.

However, this approach requires access to the host system and may introduce computational overhead, especially in resource-constrained IoT environments. Despite these limitations, host-based detection offers high accuracy and strong capability in identifying ransomware behavior compared to network-based approaches.

3.3.5 Hybrid Experimental Design

a. Preliminaries

Hybrid ransomware detection combines both network-based and host-based data to improve detection performance. The network-based approach analyzes traffic-level behavior between entities, while the host-based approach focuses on system-level activities such as processes and file operations. By integrating both perspectives, the hybrid approach aims to capture a broader range of ransomware characteristics and reduce the limitations of using a single data source.

b. Fusion Techniques

Two fusion techniques were used to construct the hybrid detection model: decision-level fusion and feature-level fusion. In the decision-level approach, the final predictions from the best-performing host-based and network-based models were combined using two methods: OR fusion and weighted fusion. The best-performing models were selected based on the highest F1-score achieved during evaluation.

The weighted fusion score was calculated as follows: $\text{Weighted Score} = (0.6 \times \text{Host Prediction}) + (0.4 \times \text{Network Prediction})$.

A slightly higher weight was assigned to the host-based prediction because host-level features provide more direct behavioral indicators of ransomware activity.

In contrast, the feature-level approach integrates the host-based and network-based features before classification. The feature sets from both domains were first aligned and then merged into a single dataset. The classification models were then retrained on this combined feature space.

c. Fusion Justification

The hybrid approach was adopted to overcome the limitations of individual detection methods. While network-based detection provides fast and scalable monitoring, it lacks visibility into internal system behavior. On the other hand, host-based detection provides detailed behavioral insights but may introduce higher overhead. By combining both data sources, the hybrid approach leverages the strengths of each method, enabling more accurate and comprehensive ransomware detection.

d. Formalizing the Hybrid Solution

The hybrid approach follows a structured ML pipeline that includes the following stages:

- **Independent Training:** The host-based and network-based datasets were preprocessed separately and used to train classification models.
- **Decision-Level Fusion:** The predictions from the best-performing models in each branch were combined using OR and weighted fusion rules.

- **Feature Alignment:** The feature sets from both domains were aligned based on the minimum common sample size after preprocessing. The records were not paired using temporal, session-based, or IP-based identifiers. Instead, the aligned subsets were concatenated column-wise to construct a unified feature space for classification. Therefore, the feature-level fusion approach in this study represents feature concatenation rather than real co-occurring host and network events.
- **Feature Merging:** The aligned features were renamed to distinguish their sources before constructing the unified hybrid datasets.
- **Model Retraining:** The classification models were retrained on the fused feature space to learn from both host and network data.
- **Prediction and Evaluation:** The hybrid models were evaluated using standard classification performance metrics.

e. Analysis of the Hybrid Approach

The hybrid approach improves ransomware detection by combining both network-level and host-level information, allowing the model to capture a wider range of malicious behavior patterns. This leads to higher detection accuracy compared to using individual approaches. However, the hybrid method introduces additional complexity in terms of data alignment and processing, and it requires higher computational resources due to the increased feature space.

One limitation of the feature-level fusion approach is that the host and network records were concatenated based on aligned sample sizes rather than real temporal or session-based pairing.

3.4 Conclusion

This chapter presented the methodology used for ransomware detection, including theoretical formalization, ML framework, and the experimental design applied in this study.

Three detection approaches were investigated: network-based, host-based, and hybrid. Each approach was formulated using the same ML framework but differed in the type of data and perspective used for detection.

The experimental design was structured to ensure a fair comparison between the approaches by using the same dataset, preprocessing steps, ML models, and evaluation metrics.

The next chapter presents the experimental results and discusses the performance of the proposed approaches.

Chapter 4: Results

4.1. Introduction

In the previous chapter, the methodology of our project was detailed including theoretical formalization, algorithm design and experimental design. In this chapter, the performance outcomes of the models will be presented based on the algorithm type and the standard evaluation metrics elaborated previously. Since we have implemented three Intrusion Detection System (IDS) types, we will have three sections in this chapter: Host-Based Results, Network-Based Results, and Hybrid Results.

4.2. Project Results

4.2.1 Host-Based Results

Results of ensemble learning models (RF, XGBoost, and CatBoost), as presented in *Table 3*, show strong and consistent performance across the evaluation metrics. For class 1 (Malicious), RF achieved a precision of 0.81, recall of 0.83, and an F1-score of 0.82, with an overall accuracy of 0.82. XGBoost reached a precision of 0.78, recall of 0.87, and an F1-score of 0.83, with an accuracy of 0.82. CatBoost also showed similar performance, with a precision of 0.76, recall of 0.92, an F1-score of 0.83, and an accuracy of 0.81.

Results of KNN, as presented in *Table 3*, achieved a precision of 0.79, recall of 0.85, and an F1-score of 0.82 for class 1, with an overall accuracy of 0.81.

Results of LR, as presented in *Table 3*, achieved a precision of 0.65, recall of 0.95, and an F1-score of 0.77 for class 1, with an overall accuracy of 0.72.

Table 3: Host-Based Detection Performance per Model (Class 1: Malicious)

Model	Class	Precision	Recall	F1-Score	Accuracy
RF	0 (Normal)	0.83	0.81	0.82	0.82
	1 (Malicious)	0.81	0.83	0.82	
XGBoost	0 (Normal)	0.87	0.78	0.82	0.82
	1 (Malicious)	0.78	0.87	0.83	
CatBoost	0 (Normal)	0.92	0.76	0.83	0.81
	1 (Malicious)	0.76	0.92	0.83	
KNN	0 (Normal)	0.85	0.79	0.82	0.81
	1 (Malicious)	0.79	0.85	0.82	
LR	0 (Normal)	0.95	0.65	0.77	0.72
	1 (Malicious)	0.65	0.95	0.77	

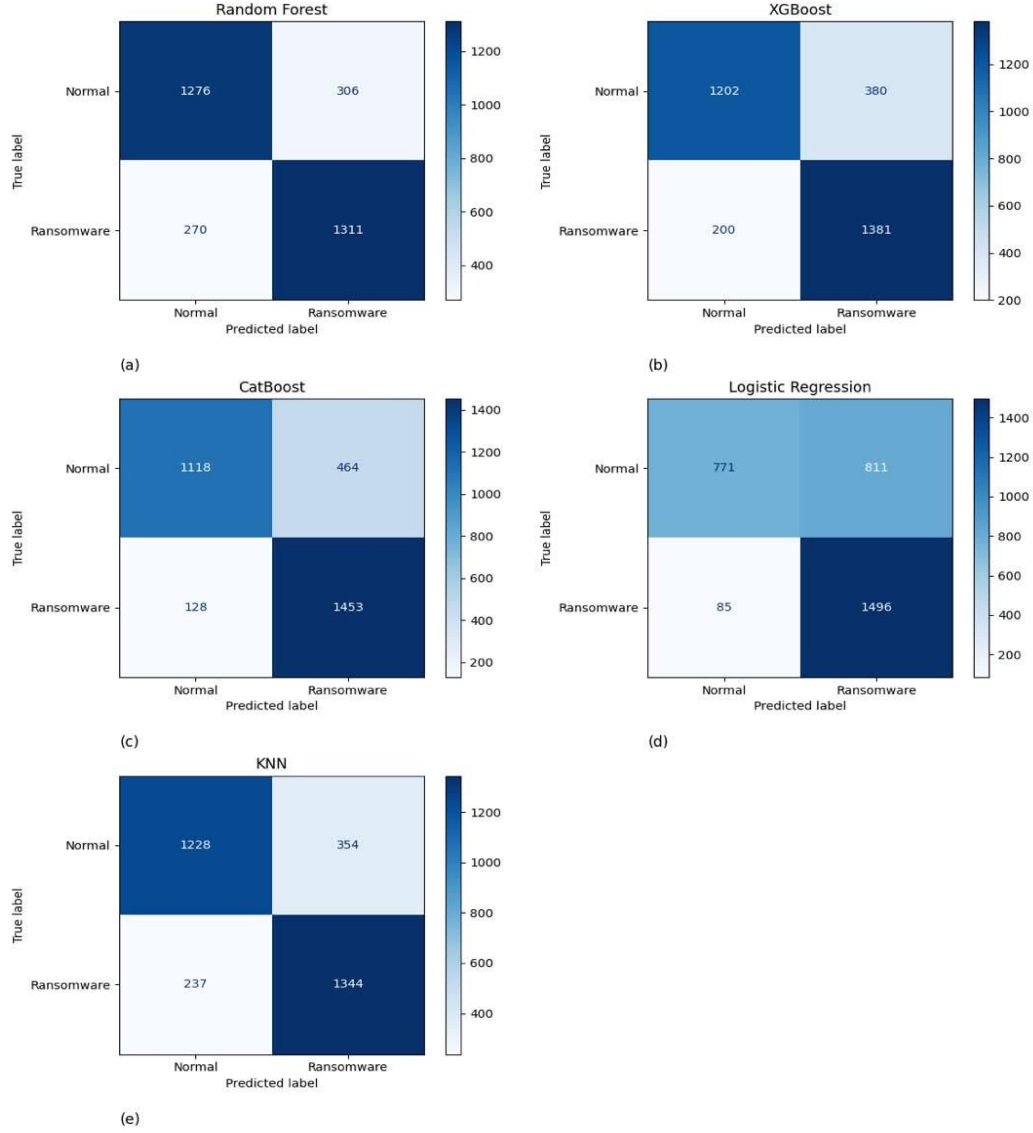


Figure 2: Confusion matrices for the five evaluated models in the Host-based experiment: (a) Random Forest, (b) XGBoost, (c) CatBoost, (d) LR, and (e) KNN

Figure 2 shows the confusion matrices of the five models. As for the tree-based ensemble models (XGBoost, RF and CatBoost), most of the classifications are clustered along the diagonal axis. RF had the least number of False Positives, 306 instances. On the other hand, the LR model had 811 False Positive cases.

4.2.2 Network-Based Results

To assess the models realistically, the test set distribution was kept the same as imbalanced as the actual. The test set comprised 173,837 samples in total, where 159,276 samples were Normal traffic (91.6%) and 14,561 samples were Ransomware (8.4%).

Results of ensemble learning models (RF, XGBoost, CatBoost) as presented in *Table 4* shows identical results across several metrics. All of them for class 1 (Ransomware), recorded precision of 0.99 except for CatBoost which scored 0.98 and a recall of 1.00 and an F1-score of 0.99. The overall accuracy was 1.00.

Results of KNN for class 1 as presented in *Table 4*, reached precision of 0.91 and a recall of 1.00 and an F1-score of 0.95 and an accuracy of 0.99.

Results of LR for class 1 as presented in *Table 4*, reached precision of 0.21 and a recall of 0.99 and an F1-score of 0.35 and an accuracy of 0.69.

Table 4: Network-Based Detection Performance per Model (Class 1: Malicious)

Model	Class	Precision	Recall	F1-Score	Accuracy
RF	0 (Normal)	1.00	1.00	1.00	1.00
	1 (Malicious)	0.99	1.00	0.99	
XGBoost	0 (Normal)	1.00	1.00	1.00	1.00
	1 (Malicious)	0.99	1.00	0.99	
CatBoost	0 (Normal)	1.00	1.00	1.00	1.00
	1 (Malicious)	0.98	1.00	0.99	
KNN	0 (Normal)	1.00	0.99	1.00	0.99
	1 (Malicious)	0.91	1.00	0.95	
LR	0 (Normal)	1.00	0.66	0.80	0.69
	1 (Malicious)	0.21	0.99	0.35	

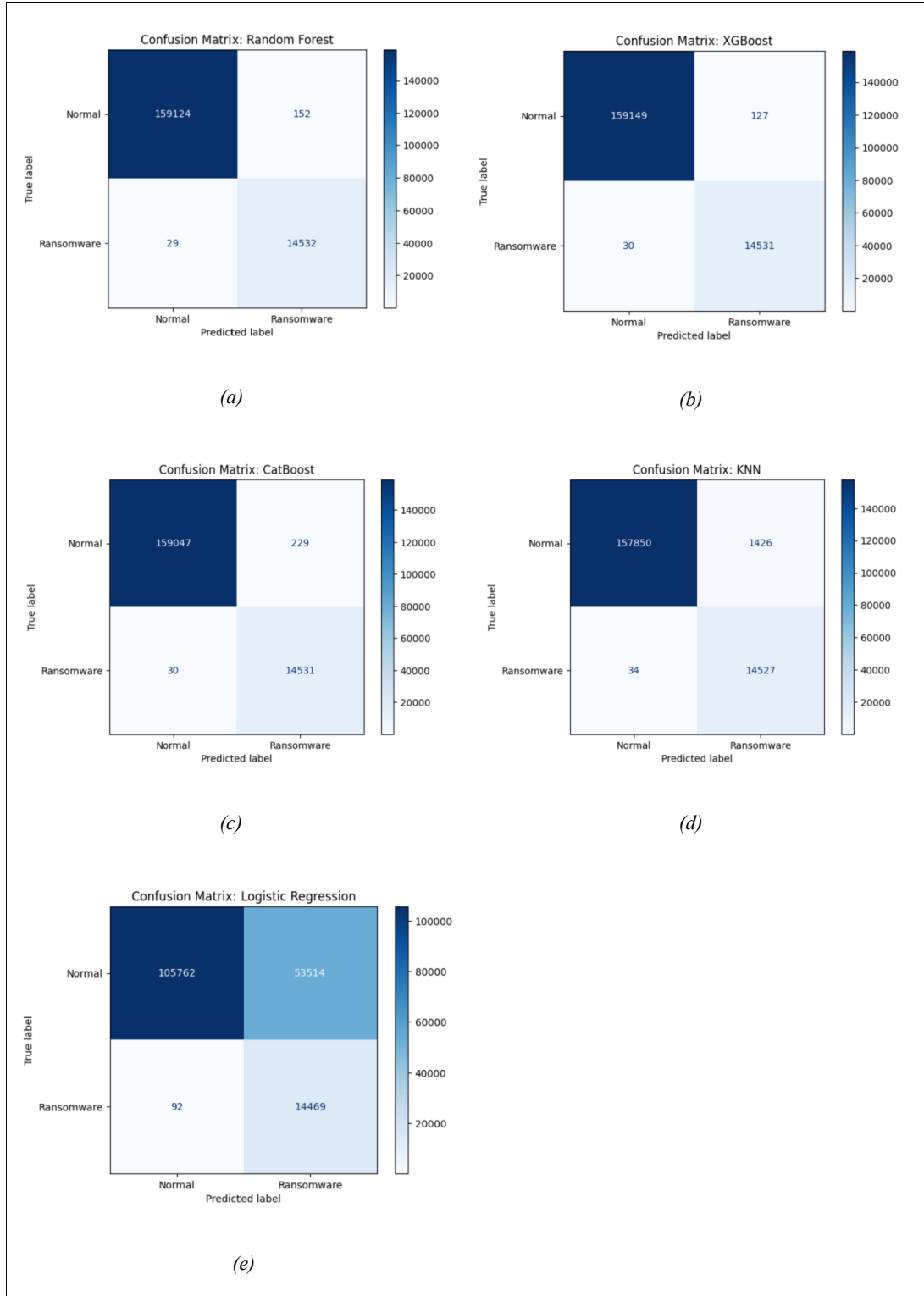


Figure 3: Confusion matrices for the five evaluated models in the Network-based experiment: (a) RF, (b) XGBoost, (c) CatBoost, (d) KNN, and (e) LR.

Figure 3 shows the confusion matrices of the five models. As for the tree-based ensemble models (XGBoost, RF and CatBoost), most of the classifications are clustered along the diagonal axis. XGBoost had the least number of False Positives, 127 instances. On the other

hand, the LR model had 53,514 False Positive cases. The matrices indicate that the test set distribution was 159,276 normal samples and 14,561 ransomware samples.

4.2.3 Hybrid Results

As shown in *Table 5*, the hybrid detection stage was evaluated using both decision-level fusion and feature-level fusion based on the performance of class 1 (Malicious). In the decision-level approach, OR fusion achieved a precision of 0.75, recall of 0.87, F1-score of 0.80, and an overall accuracy of 0.79. The weighted fusion method slightly improved the malicious-class detection, reaching a precision of 0.79, recall of 0.85, F1-score of 0.82, and an accuracy of 0.81.

For feature-level fusion, the classifiers were retrained on the fused host and network feature space. The best results for class 1 were achieved by XGBoost and CatBoost, where both models achieved perfect performance with precision, recall, and F1-score values of 1.00, alongside an overall accuracy of 1.00. These results clearly outperform the decision-level fusion methods.

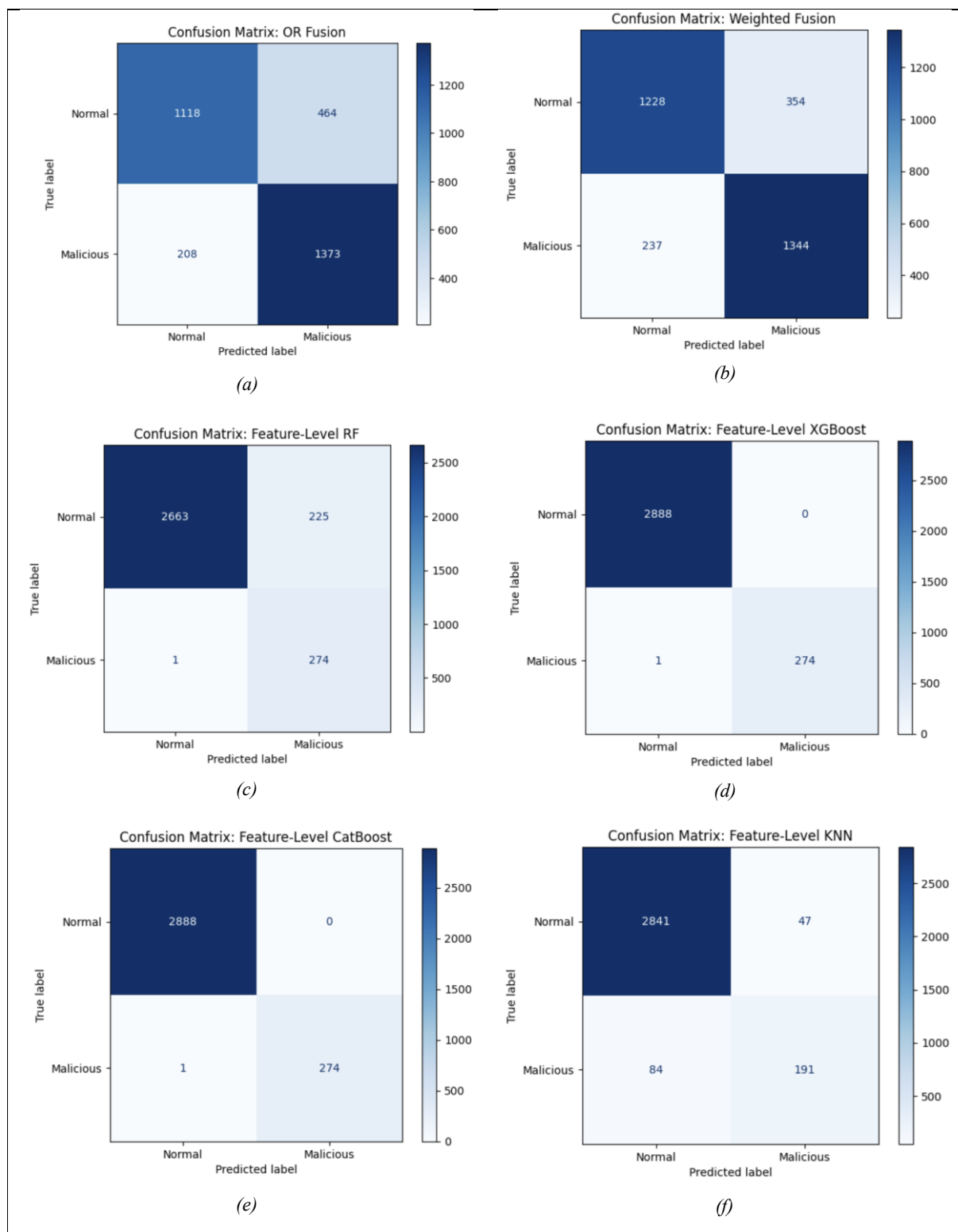
While the training data was balanced during preprocessing, the test set distribution was intentionally kept imbalanced to reflect more realistic evaluation conditions. In addition, confusion matrices were used to analyze the classification performance and reduce the possibility of misleading results caused by class imbalance. The near-perfect performance achieved by XGBoost and CatBoost may be related to the structured nature of the fused feature space and the strong separability between the Normal and Malicious classes within the evaluated dataset.

Among the remaining feature-level models, KNN also showed strong detection capability with a precision of 0.80, recall of 0.69, F1-score of 0.74, and an accuracy of 0.96.

Table 2: Hybrid Detection Performance per Model (Class 1: Malicious)

Fusion Type	Model	Class	Precision	Recall	F1-Score	Accuracy
Decision-Level	OR Fusion	0 (Normal)	0.84	0.71	0.77	0.79
		1 (Malicious)	0.75	0.87	0.80	
Decision-Level	Weighted Fusion	0 (Normal)	0.84	0.74	0.81	

		1 (Malicious)	0.79	0.85	0.82	0.81
Feature-Level	RF	0 (Normal)	1.00	0.92	0.96	0.93
		1 (Malicious)	0.55	1.00	0.71	
Feature-Level	XGBoost	0 (Normal)	1.00	1.00	1.00	1.00
		1 (Malicious)	1.00	1.00	1.00	
Feature-Level	CatBoost	0 (Normal)	1.00	1.00	1.00	1.00
		1 (Malicious)	1.00	1.00	1.00	
Feature-Level	KNN	0 (Normal)	1.00	0.66	0.80	0.96
		1 (Malicious)	0.80	0.69	0.74	
Feature-Level	LR	0 (Normal)	0.97	0.98	0.98	0.69
		1 (Malicious)	0.22	0.99	0.36	



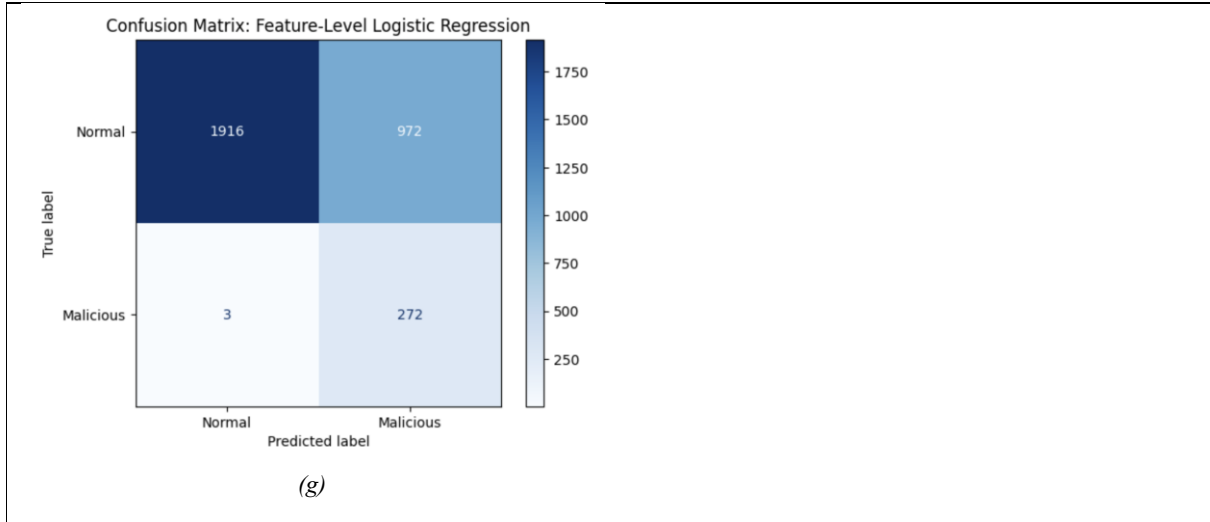


Figure 4: Confusion matrices for the hybrid detection models using decision-level and feature-level fusion approaches: (a) OR Fusion, (b) Weighted Fusion, (c) Feature-Level RF, (d) Feature-Level XGBoost, (e) Feature-Level CatBoost, (f) Feature-Level KNN, and (g) Feature-Level LR.

Figure 4 shows the confusion matrices for hybrid fusion models. The feature-level models achieved the best classification performance with very low misclassification rates, especially XGBoost and CatBoost. On the other hand, KNN and LR showed lower performance due to a higher number of false predictions. The decision-level fusion methods achieved balanced results but produced more false positives compared to the feature-level fusion models.

Chapter 5: Discussion

5.1 Introduction

In the previous chapter we presented the performance results of the ML models across the three proposed approaches: host-based, network-based, and hybrid using standard evaluation metrics.

The focus of this chapter is to discuss the results and to interpret them critically in relation to the project objectives and the limitations elaborated earlier in chapter two, that means this chapter will address how our research addresses the key gaps in the literature and how it is contributing to field of ransomware detection specifically in IoT environments.

5.2 Discussion of the Results

As identified previously in the literature review chapter, many existing studies depend on a single source of evidence such as network-based only [19] or host-based only [17] which puts a limit over the ability to detect ransomware effectively. Also, many studies relied on heavy approaches and models such as GRU-TCN [18] which makes it unsuitable for IoT's limited resources environments. As well as using simulated IoT environments rather than a realistic one [12].

The proposed approach addresses these limitations by integrating host-based and network-based data within a unified ML framework using a dataset collected from a realistic IoT testbed environment..

1. Host-Based discussion

The results demonstrated a stable performance, especially with ensemble learning models (RF, XGBoost, CatBoost). These results are aligning with previous works such as L-IDS [12], which demonstrate that ML models can achieve strong detection performance in this context.

Yet, relying on Host-Based detection alone showed a certain limitation in both this research and in previous studies. For example, the high recall but lower precision observed in LR means that there might be a tendency towards false positives which supports the observation in the

literature that host-level detection alone may over-detect anomalies due to limited contextual awareness.

2. Network-Based discussion

The ensemble learning models demonstrated optimal performance, especially for XGBoost, RF and CatBoost. These results are consistent with the existing literature [19] such as on the IoT-NIDS systems. The effectiveness of the network traffic features to detect malicious traffic are evident. To provide a more accurate evaluation the models were run on an imbalanced data set which is closer to the real-world environment where the normal traffic is substantially more compared to ransomware.

Another critical outcome of this study is the comparison between ensemble and linear models (such as LR). The trees-based models demonstrated good recall and precision values for both classes. In the case of LR an accuracy of 0.69 and a very low precision of 0.21 were achieved for the ransomware class. These values can be attributed to the presence of normal traffic in the test set (91.6%). A crucial metric to consider is that the LR obtained bad results, as the number of False Positives obtained by LR model is extremely high. In its confusion matrix there are 53,514 instances of False Positives. This demonstrates that the classification problem is complex and the high values of accuracy (1.00) for XGBoost and RF have been attained due to the models' ability to consider the non-linear relations, rather than by a majority class assumption.

A transparent view of the reliability of the system can be observed in class 0 (Normal) metrics and the respective confusion matrices. The XGBoost confusion matrix presents only 127 False Positives over the total 159,276 normal traffic. A high specificity of the system in detecting the normal traffic is crucial in the IoT context, in order to avoid unnecessary service interruption. This work was executed on an imbalanced test set and it has proved that ensemble of models-based solutions can maintain robust decision boundaries without the majority class assumption influencing the final results.

Although these results are highly promising, one should be aware of the limitation of only having network-based detection systems. As already seen from the literature, network traffic analysis cannot always be useful if the amount of traffic is relatively low or if the attack is in its silent, early stage. The present work has obtained good results on the ToN-IoT data set, however feature selection and appropriate algorithm selection remain important. Future solutions will probably incorporate host-based indicators for increased security, building upon the real time network based ransomware detection systems.

3. Hybrid discussion

This approach is providing the most significant contribution in this research, by combining both host and network features, the proposed model directly addresses the limitation of single source of evidence identified in several studies [19].

The feature-level fusion results, specifically with the feature-level fusion using XGBoost and CatBoost achieved an F1-score of 1.00, representing an absolute improvement of 0.18 points over the best decision-level OR-fusion ($F1 = 0.82$). It also outperforms the reported results of similar studies such as EPS-Ran [18] which achieved 96.98% and IoT-NIDS [19] which achieved 96.98%.

Furthermore, the hybrid feature-level outperformed standalone host-based models where the best F1-score was 0.83 (XGBoost/CatBoost), showing a gain of 0.17 points through feature integration. These results indicate that merging raw features spaces enable the model to detect more complex ransomware patterns.

These observations are aligned with previous studies such as the approach of CrossSentry [14] which is a cross-layer approach to ransomware detection. However, unlike CrossSentry, which was mainly tested on a few numbers of devices, the proposed approach demonstrates strong performance using a more ML framework.

Overall, the results show that using a multi-source for detection is outperforming relying on a single source.

4. Suggestions for future work

Despite the promising outcomes, certain limitations remain, specifically regarding early-stage detection and real-world IoT deployment which will be further elaborated in the next chapter.

5.3 Conclusion

A detailed interpretation of the results was presented in this chapter, focusing on the relationship between this research and previous studies. The observations demonstrate that multi-source detection approaches provide stronger and more realistic performance compared to relying on a single detection source. Furthermore, the results show that ML models can achieve strong ransomware detection performance across different detection approaches.

These observations contribute to the field of IoT security effectively, as they provide a practical, effective approach for ransomware detection. It is also contributing to the advancement of intelligent IDS in Computer Science.

The next chapter will summarize the overall work and will discuss future work directions.

Chapter 6: Conclusion and Future Work

6.1 Introduction

In the previous chapter, the results were critically interpreted, discussed, and examined in relation to the research objectives and existing literature. The observations showcased the effectiveness of the proposed model in detecting ransomware in IoT environments.

In this chapter, the overall conclusion of this research project, summarizing the main findings, examining how the proposed approach meets the research objectives, identifying the limitations, and outlines the potential direction for future work.

6.2 Contributions

This project contributes to the field of IoT security and specifically in ransomware detection in several key aspects:

- Developed three ransomware detection models based on three different perspectives: Host-Based, Network-Based, and Hybrid.
- Addressed the key limitation of relying on a single source of evidence by merging both host and network level features within a single framework.
- All these contributions are collectively supporting that is suitable for resource-constrained IoT environments.
- Conducted a comprehensive experimental evaluation by comparing several classifiers across different detection approaches.
- Showcased that hybrid feature-level fusion improves the overall detection performance compared to standalone models.
- Provided a detailed analysis with regard to previous studies and addressed some of the key gaps found in the literature.

All these contributions are collectively supporting the development of more effective IDS for IoT environments.

6.3 Limitations

- **Offline Evaluation:** The proposed models were evaluated using pre-collected, offline datasets, the system was not tested in a real-time, live environment.
- **Controlled Environment:** The near-perfect detection results are attributed to the controlled nature and well-structured nature of the laboratory test-bed environment from which the data was derived.
- **Dataset Specificity:** This research focused specifically on the ToN-IoT dataset, the generalizability of these results and findings to other IoT architectures or other ransomware families not represented in the used dataset may require further experimental validation.
- **Lack of Resource Evaluation:** This study did not include quantitative measurements related to computational resource usage, such as CPU utilization, memory consumption, or inference time.

6.4 Future works

Several improvements and directions can be explored to further enhance the proposed model:

- Training the model using real IoT devices.
- Implementing early-stage detection features to be part of the proposed model so that the detection starts before encryption begins.
- Exploring more advanced optimization techniques and feature selection to improve efficiency.
- Evaluating Time-to-Detect (TTD) in real-time scenarios.
- Measuring system performance overhead, including CPU usage, memory consumption, and latency in IoT environments.

6.5 Conclusion

This project addressed the challenge of ransomware detection in IoT resource-constrained environment, where traditional approaches usually used only single source of evidence or computationally heavy models.

The main objective of this study was to investigate ransomware detection in IoT environments using ML approaches. This objective was achieved by designing and evaluating host-based, network-based, and hybrid detection models..

The results showed that hybrid feature-level approach provided the most effective and accurate performance, overcoming the limitation of single-source detection.

Despite these achievements, the study has its limitations such as that the early-stage detection remains an open challenge and that the models were evaluated in a controlled environment.

Overall, the proposed approach demonstrates a practical and effective solution for ransomware detection in IoT environments.

Appendices

Appendix 1: Project Source Code Link

<https://github.com/lulwah-mohammed/GP-F31/tree/main>

Appendix 2: Project Poster



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